

PROJECT HOSPITAL DATABASE



KING KHALID UNIVERSITY
College of Science and Arts-KH
DEPARTMENT OF COMPUTER SCIENCE



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INTRODUCTION

A mini-project of the Hospital Management Database

The project talks about the way to save the data of the hospital administration and its employees, in addition to customer data, facilitating access and linking it. There are 9 tables, each table that saves part of the important and related information to another table. Table 1 Administration :It contains the personal information of the administration + supervision and follow-up of work and staff and performance evaluation. Table 2 Stuff : It contains the personal information of the rest of the employees such as receptionists, cleaners and security. Table 3 Reception: Patient information and data for booking. Table 4 Patients: Patients' Personal Information. Table 5 Departments : It contains the information of the departments for doctors with patients for each department. Table 6 Doctors: Doctors' Personal Information. Table 7 Nurses: Nurses' Personal Information. Table 8 rooms: A table dedicated to knowing the number of the room, its type and its condition, whether it is empty or not, and whether there is a companion for the owner of the room or not. Finally Table 9 bill: Table for calculating the value of the patient's cost during his stay.

ENTITIES

- **Stuff :** Personnel Persons working in the hospital in general, such as security, cleaners and other workers
- **Doctor :** The person who helps the sick and diagnoses the disease
- **Nurse :** The person responsible for nursing care in the various stages of the patient's presence
- **Administration :** The persons authorized to control, administer, and supervise public health systems and health care systems
- **Room :** A place responsible for providing medical and surgical care to incoming patients
- **Patient :** Any person receiving medical attention, care or treatment.
- **Department :** The medical departments in the hospital
- **Bill :** Hospital fees according to the patient's illness
- **Reception :** Receiving patients and their families, reassuring them, and answering all their questions and inquiries.

STUUF

Columns name	Description	Type	Length
Stuff_id(PK)	Stuff id	numder	4
Job_title	type of job	varchar2	30
Name	Stuff name	varchar2	30
Age	Stuff age	numder	2
Gender	Stuff gender	varchar2	10
Address	Stuff address	varchar2	30
Phone_no	Stuff phone	numder	10
Salary	Stuff salary	numder	6
Work_hour	Stuff work hour	numder	2
Email	The email	varchar2	30
Nationality	Stuff nationality	varchar2	30
Status	Stuff status	varchar2	10

DOCTOR

Columns name	Description	Type	Length
Doctor_id(PK)	Doctor id	numder	4
Name	Doctor name	varchar2	30
Age	Doctor age	numder	2
Gender	Doctor gender	varchar2	10
Address	Doctor address	varchar2	30
Phone_no	Doctor phone	numder	10
Salary	Doctor salary	numder	6
Section	Doctor section	varchar2	30
Work_hour	Doctor work hour	numder	2
Email	The email	varchar2	30
Nationality	Doctor nationality	varchar2	30
Experience	Doctor experience	numder	2
Status	Doctor status	varchar2	10
Graduation_uni	Doctor graduation	varchar2	50

NURSE

Columns name	Description	Type	Length
Nurse_id(PK)	Nurse id	numder	4
Name	Nurse name	varchar2	30
Age	Nurse age	numder	2
Gender	Nurse gender	varchar2	10
Address	Nurse address	varchar2	30
Phone_no	Nurse phone	numder	10
Salary	Nurse salary	numder	6
Work_hour	Nurse work hour	numder	2
Email	The email	varchar2	30
Nationality	Nurse nationality	varchar2	30
Experience	Nurse experience	numder	2
Status	Nurse status	varchar2	10
Graduation_uni	Nurse graduation	varchar2	50

ADMINISTRATION

Columns name	Description	Type	Length
Administration_id(PK)	Administration id	numder	4
Name	Administration name	varchar2	30
Age	Administration age	numder	2
Gender	Administration gender	varchar2	10
Address	Administration address	varchar2	30
Phone_no	Administration phone	numder	10
Salary	Administration salary	numder	6
Work_hour	Administration work hour	numder	2
Email	The email	varchar2	30
Nationality	Administration nationality	varchar2	30
Status	Administration status	varchar2	10
Graduation_uni	Administration graduation	varchar2	50
Doctor_id(FK)	Doctor id	numde	4
Nurse_id(FK)	Nurse id	numde	4
Stuff_id(FK)	Stuff id	numde	4

ROOM

Columns name	Description	Type	Length
Room_no(PK)	Room number	number	4
Room_type	Room type	varchar2	30
Situation	room situation	varchar2	30
Facilities	facilities name	varchar2	30

PATIENT

Columns name	Description	Type	Length
Patient_id(PK)	Patient id	numder	4
Name	Patient name	varchar2	30
Age	Patient age	numder	2
Gender	Patient gender	varchar2	10
Address	Patient address	varchar2	30
Phone_no	Patient phone	numder	10
Email	The email	varchar2	30
Doctor_id(FK)	Doctor id	numder	4
Nurse_id(FK)	Nurse id	numder	4
Room_no(FK)	Room number	numder	4

DEPARTMENT

Columns name	Description	Type	Length
Dept_id(PK)	Department id	numder	4
Dept_name	Department name	varchar2	30
Room_no(FK)	Room number	numder	4
Patient_id(FK)	Patient id	numder	4
Doctor_id(FK)	Doctor id	numder	4

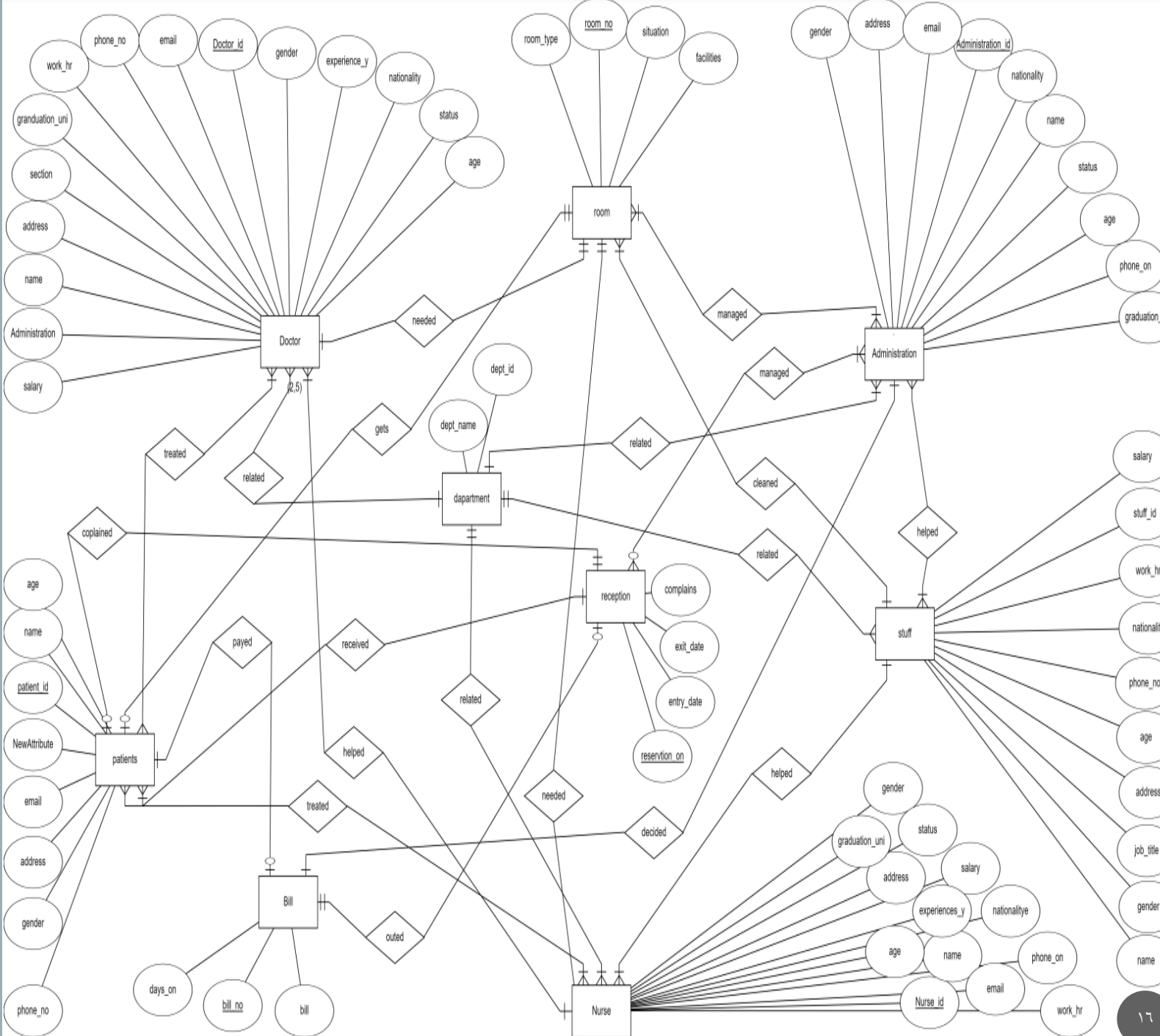
BILL

Columns name	Description	Type	Length
Bill_no(PK)	Patient id number	number	10
Patient id(FK)	Patient id	number	4
Doctor_id(FK)	Doctor id	number	4
Room_no(FK)	Room number	number	4
Days_no	number of days spent	number	10
Dept_id(FK)	Department id	number	4
Bill	The patient's bill	number	10

RESERVATION

Columns name	Description	Type	Length
Reservation_no(PK)	Reception number	number	4
Patient_id(FK)	Patient id	number	4
Date_entry	Date of entry	number	10
Date_exit	Date of exit	number	10
complaints	The problem	varchar2	70
Room_no(FK)	Room number	number	4
Dept_id(FK)	Department id	number	4

THE ER-DIAGRAM



CREATE AND INSERT ADMINISTRATION

```
SQL> create table administration (Administrat_id number ( 4) primary key, Name varchar (30 ), Age number ( 2), Gender varchar (10 ), Address varchar (30 ), Phone_no number (10 ) not null, Salary number ( 6) not null, Work_hour number (2 ), Email varchar (30), Nationat varchar (30 ), Status varchar (10 ), Gradua_uni varchar (30 ), doctor_id number (4 )not null, Nurse_id number (4)not null, Stuff_id number ( 4) not null);
```

```
SQL> ALTER TABLE administration ADD CONSTRAINT fk_nu FOREIGN KEY (Nurse_id) REFERENCES Nures(Nurse_id);
```

```
SQL> ALTER TABLE administration ADD CONSTRAINT fk_st FOREIGN KEY (Stuff_id) REFERENCES stuff (Stuff_id);
```

```
ALTER TABLE administration ADD CONSTRAINT fk_do FOREIGN KEY (doctor_id) REFERENCES doctor(doctor_id);
```

```
SQL> insert into Administration values(5098,'Sara',35,'female','ABha',0567782340,12000,7,'sara45@gmail.com','Saudi','married','Princess Nourah University',1970,2234,1133);
```

```
SQL> insert into Administration values(5123,'Ali',28,'male','Riyadh',0567782110,10000,7,'ali9@gmail.com','Saudi','married','King Abdulaziz University',1245,2209,2341);
```

```
SQL> insert into Administration values(5166,'Mohand',39,'male','Qassim',0567784321,20000,9,'mo7@gmail.com',  
'Saudi','married','Imam Mohammad University',1576,2243,9837);
```

```
SQL> insert into Administration values(5567,'Saleh',25,'male','Najran',0563094321,9000,4,'ssaleh7@gmail.com',  
Saudi,'single','Umm AlQura University',1880,2876,5654);
```

```
SQL> insert into Administration  
values(5437,'Amal',29,'female','AlJouf',0563094008,15000,8,'amal@gmail.com','Saudi','married','King Khalid  
University',1080,2004,1118);
```

```
SQL> insert into Administration values(5020,'Masoud',41,'male','Dammam',0563097608,25000,7,'mas8@gmail.c  
om','Saudi','married','Taibah University',1139,2010,1654);
```

```
SQL> insert into Administration  
values(5419,'Fatimah',22,'female','AlBaha',0563097809,13000,9,'fatom@gmail.com','Saudi','single','King Khalid  
University',1742,2023,8900);
```

```
SQL> insert into Administration values(5451,'Salwa',36,'female','Makkah',0563000608,31000,9,'salo@gmail.com',  
'Saudi','married','Taibah University',1623,2809,0900);
```

```
SQL> insert into Administration values(5600,'Hajer',32,'female','ABha',0567782800,25000,8,'hajaer9@gmail.com',  
'Saudi','married','Princess Nourah University',1552,2870,8000);
```

```
SQL> insert into Administration values(5117,'Mohammed',30,'male','Najran',0563780321,14000,6,'mohha7@gmail.com',  
'Saudi','single','Umm AlQura University',1980,2016,8904);
```

```

SQL> create table administration (Administrat_id number ( 4) primary key, Name varchar (30 ), Age
2   number ( 2), Gender varchar (10 ), Address varchar (30 ), Phone_no number (10 ) not null, Salary
3   number ( 6) not null, Work_hour number (2 ), Email varchar (30), National varchar (30 ), Status
4   varchar (10 ), Gradua_uni varchar (30 ), doctor_id number (4 )not null, Nurse_id number (4)not null, Stuff_id number ( 4) not null);

Table created.

SQL> ALTER TABLE administration ADD CONSTRAINT fk_nu FOREIGN KEY (Nurse_id) REFERENCES Nures(Nurse_id);

Table altered.

SQL> ALTER TABLE administration ADD CONSTRAINT fk_st FOREIGN KEY (Stuff_id) REFERENCES stuff (Stuff_id);

Table altered.

SQL> insert into Administration values(5098,'Sara',35,'female','Abha',0567782340,12000,7,'sara45@gmail.com','Saudi','married','Princess Nourah University',1970,2234,1133);

1 row created.

SQL> insert into Administration values(5123,'Ali',28,'male','Riyadh',0567782110,10000,7,'ali19@gmail.com','Saudi','married','King Abdulaziz University',1245,2209,2341);

1 row created.

SQL> insert into Administration values(5166,'Mohand',39,'male','Qassim',0567784321,20000,9,'mo7@gmail.com','Saudi','married','Imam Mohammad University',1576,2243,9837);

1 row created.

SQL> insert into Administration values(5567,'Saleh',25,'male','Najran',0563094321,9000,4,'ssaleh7@gmail.com','Saudi','single','Umm AlQura University',1880,2876,5654);

1 row created.

SQL> insert into Administration values(5437,'Amal',29,'female','AlJouf',0563094008,15000,8,'amal@gmail.com','Saudi','married','King Khalid University',1080,2004,1118);

1 row created.

SQL> insert into Administration values(5020,'Masoud',41,'male','Dammam',0563097608,25000,7,'mas8@gmail.com','Saudi','married','Taibah University', 1139,2010,1654);

1 row created.

SQL> insert into Administration values(5419,'Fatinah',22,'female','AlBaha',0563097809,13000,9,'fatou@gmail.com','Saudi','single','King Khalid University',1742,2023,8900);

1 row created.

SQL> insert into Administration values(5451,'Salwa',36,'female','Makkah',0563000608,31000,9,'salo@gmail.com','Saudi','married','Taibah University',1623,2809,0900);

1 row created.

SQL> insert into Administration values(5600,'Hajer',32,'female','Abha',0567782800,25000,8,'hajaer9@gmail.com','Saudi','married','Princess Nourah University',1552,2870,8000);

1 row created.

```

```
SQL> select * from administration;
```

ADMINISTRAT UNIT	ID NAME	DOCTOR_ID	NURSE_ID	STUFF_ID	AGE	GENDER	ADDRESS	PHONE_NO	SALARY	WORK_HOUR	EMAIL	NATIONAT	STATUS	GRADUA
ss Hourah University	5098 Sara	1070	2234	1133	35	female	ABha	567782340	12000	7	sara45@gmail.com	Saudi	married	Prince
ndulaziz University	5123 Ali				28	male	Riyadh	567782110	10000	7	ali9@gmail.com	Saudi	married	King A
phammad University	5106 Mohand		1245	2209	39	male	Qassim	567784321	20000	9	mo7@gmail.com	Saudi	married	Imam M
Qura University	5567 Saleh		1576	2243	25	male	Najran	563894321	9000	4	ssaleh7@gmail.com	Saudi	single	Umm Al
halid University	5437 Amal		1880	2876	29	female	AlJouf	563094008	15000	8	amel@gmail.com	Saudi	married	King K
University	5020 Masoud		1139	2010	41	male	Dammam	563897608	25000	7	mas8@gmail.c om	Saudi	married	Taibah
halid University	5419 Fatimah		1742	2023	22	female	AlBaha	563897809	13000	9	fatom@gmail.com	Saudi	single	King K
University	5451 Salwa		1623	2809	36	female	Makkah	563800608	31000	9	salo@gmail.com	Saudi	married	Taibah
ss Hourah University	5600 Hajer	1552	2870	8000	32	female	ABha	567782800	25000	8	hajaer9@gmail.com	Saudi	married	Prince
Qura University	5117 Mohammed	1980	2016	8904	38	male	Najran	563780321	14000	6	mohha7@gmail.com	Saudi	single	Umm Al

10 rows selected.

```
SQL>
```

CREATE AND INSERT STUFF

```
SQL> create table stuff ( Stuff_id number ( 4) primary key, Jop_title varchar (30 ), Name varchar (30 ), Age number (2 ), Gender varchar (10 ), Address varchar ( 30), Phone_no number (10 ) not null, Salary number ( 6) not null, Work_hour number ( 2), Email varchar (30 ) , Nationality varchar (30) , Status varchar (10 ));
```

```
SQL> Insert into stuff values ( 1133,'secretary','ahmed',23,'male','abha',05147878,1900,9,'hhds@gamill.com','saudi','unmarried');
```

```
SQL> Insert into stuff values ( 2341, ' secretary','sara',22, 'famle','jedah',05765558,1900,9,'ssas@gamill.com','pahistan','marrird');
```

```
SQL> Insert into stuff values ( 9837,'rossy', ' secretary',19,'famle ','abha',055678784,1900,9, 'hrrrr@gamill.com','indonesi','unmarried');
```

```
SQL> Insert into stuff values ( 5654,'clean work ','snda',40,'famle ','kasiem',051478765,1900, 11,'h777hs@gamill.com','filippinres','married');
```

```
SQL> Insert into stuff values ( 1118,'clean work ','nourh', 19,'famle','najran',053336638,1900,11,'looo@gamill.com','filippinres','unmarried');
```

```
SQL> Insert into stuff values ( 1654,'clean work ','rayan',53,'male',' tabouk ',055557624,1900,33,'2232j@gamill.com','indian','marrird');
```

```
SQL> Insert into stuff values ( 8900,'secretary','khalid',30,'male',' tabouk ',05147665,1900,12,'saaaa@gamill.com','saudi','marrid');
```

```
SQL> Insert into stuff values ( 0900,'secretary','saad',38,'male','bridah',05144465,5900,12,'saaoa@gamill.com','saudi','unmarrid');
```

```
SQL> Insert into stuff values ( 8000,'pharmacits','smeer',30,'male','rayidh',05147665,1190,12,'saaaa@gamill.com','eqypt','marrid');
```

```
SQL> Insert into stuff values ( 8904,'pharmacits','sultan',30,'male','abha',05147665,1900,12,'saaaa@gamill.com','saudi','marrid');
```

```
Run SQL Command Line
SQL> create table stuff ( Stuff_id number ( 4 ) primary key, Jop_title varchar ( 30 ), Name varchar ( 30 ), Age number ( 2 ), Gender varchar ( 10 ), Address varchar ( 30 ), Phone_no number ( 10 ) not null, Salary number ( 6 ) not null, Work_hour number ( 2 ), Email varchar ( 30 ), Nationality varchar ( 30 ), Status varchar ( 10 ));
Table created.
SQL> Insert into stuff values ( 1133,'secretary','ahmed',23,'male','abha',05147878,1900,9,'hhds@gamill.com','saudi','unmarried');
1 row created.
SQL> Insert into stuff values ( 2341, ' secretary','sara',22, 'femle','jedah',05765558,1900,9,'ssas@gamill.com','pahistan','marrird');
1 row created.
SQL> Insert into stuff values ( 9837,'rossy', ' secretary',19,'femle ','abha',055678784,1900,9, 'hrrrr@gamill.com','indonesi','unmarried');
1 row created.
SQL> Insert into stuff values ( 5654,'clean work ','snda',40,'femle ','kasien',051478765,1900, 11,'h777hs@gamill.com','filippinres','married');
1 row created.
SQL> Insert into stuff values ( 1118, 'clean work ','nourh', 19,'femle','najran',053336638,1900, 11,'looo@gamill.com','filippinres','unmarried');
1 row created.
SQL> Insert into stuff values ( 1654, 'clean work ','rayan',53,'male',' tabouk ',055557624,1900,33, '2232j@gamill.com','indian','marrird');
1 row created.
SQL> Insert into stuff values ( 8900,'secretary','khalid',30,'male',' tabouk ',05147665,1900,12, 'saaaa@gamill.com','saudi','marrid');
1 row created.
SQL> Insert into stuff values ( 0900,'secretary','saad',38,'male','bridah',05144465,5900,12, 'saaoa@gamill.com','saudi','unmarrid');
1 row created.
SQL> Insert into stuff values ( 8000, 'pharmacits','sneer',30,'male','rayidh',05147665,1190,12, 'saaaa@gamill.com','eqypt','marrid');
1 row created.
SQL> Insert into stuff values ( 8904,'pharmacits','sultan',30,'male','abha',05147665,1900,12, 'saaaa@gamill.com','saudi','marrid');
1 row created.
SQL> select * from stuff;
STUFF_ID JOP_TITLE NAME AGE GENDER ADDRESS PHONE_NO SALARY WORK_HOUR EMAIL NATIONALITY
STATUS
```


row created.

SQL> select * from stuff;

STUFF_ID	JOB_TITLE STATUS	NAME	AGE	GENDER	ADDRESS	PHONE_NO	SALARY	WORK_HOUR	EMAIL	NATIONALITY
1133	secretary unmarried	ahmed	23	male	abha	5147878	1900	9	hhds@gamil.com	saudi
2341	secretary married	sara	22	female	jedah	5765558	1900	9	ssas@gamil.com	pakistan
9837	bossy unmarried	secretary	19	female	abha	55678784	1900	9	hrrrr@gamil.com	indonesi
5654	clean work married	snda	40	female	kasien	51478765	1900	11	h777hs@gamil.com	filippinres
1118	clean work unmarried	nourh	19	female	najran	53336638	1900	11	looo@gamil.com	filippinres
1654	clean work married	rayan	53	male	tabouk	55557624	1900	33	2232j@gamil.com	indian
8900	secretary married	khalid	30	male	tabouk	5147665	1900	12	saaaa@gamil.com	saudi
900	secretary unmarried	saad	38	male	bridah	5144465	5900	12	saaco@gamil.com	saudi
8000	pharmacits married	smeer	30	male	rayidh	5147665	1190	12	saaaa@gamil.com	egypt
8904	pharmacits married	sultan	30	male	abha	5147665	1900	12	saaaa@gamil.com	saudi

9 rows selected.

CREATE AND INSERT PATIENT

```
create table patients ( Patient_id number (4 ) primary key, Name varchar ( 30), Age number (2 ), Gender varchar (10 ), Address varchar (30 ), Phone_no number (10 ) not null, Email varchar ( 30), doctor_id number (4 )not null, Nurse_id number (4)not null ,room_no number ( 4)not null);
```

```
.SQL> ALTER TABLE patients ADD CONSTRAINT fk_dp FOREIGN KEY (doctor_id) REFERENCES doctor(doctor_id);
```

```
SQL> ALTER TABLE patients ADD CONSTRAINT fk_np FOREIGN KEY (Nurse_id) REFERENCES Nures(Nurse_id);
```

```
SQL> ALTER TABLE patients ADD CONSTRAINT fk_ro FOREIGN KEY (room_no) REFERENCES room(room_no);
```

```
SQL> Insert into patients values (9777,'reem',1,'famel','riyadh',051111189, 'm88@outlook.sa',1245, 2209,911);
```

SQL> Insert into patients values (9119,'hasen',19,'male','makah',08765489, 'ccx8@gamil.sa',1576, 2243,965);

SQL> Insert into patients values (1139,'nada',10,'famel','abha',05611129, 'aas88@outlook.sa',1880, 2876, 950);

SQL> Insert into patients values (9009,'kalid',90,'male','Najran',05836889, 'a388@ gamil.sa',1080, 2004,977);

SQL> Insert into patients values (9111,'njla',40,'famel','yunba',01137333, 'a2188@outlook.sa',1139, 2010,0968);

SQL> Insert into patients values (1129,'sultan',28,'male','hail',0561239, '09m88@outlook.sa',1742, 2023,0912);

SQL> Insert into patients values (1219,'nada',10,'famel','abha',05611129, 'aas88@outlook.sa',1623, 2809,0980);

SQL> Insert into patients values (2129,'njla',40,'famel','yunba',01137333, ' a2188@outlook.sa',1980, 2016,0901);

SQL> Insert into patients values (8976,'sultan',28,'male','hail',0561239, '09m88@outlook.sa' ,1876,2094,0951);

```

QL> create table patients ( Patient_id number (4 ) primary key, Name varchar ( 30), Age number (2 ), Gender varchar (10 ), Address varchar (30 ),
2 Phone_no number (10 ) not null, Email varchar ( 30), doctor_id number (4 )not null, Nurse_id number (4)not null ,room_no number ( 4)not null);

able created.

QL> ALTER TABLE patients ADD CONSTRAINT fk_dp FOREIGN KEY (doctor_id) REFERENCES doctor(doctor_id);

able altered.

QL> ALTER TABLE patients ADD CONSTRAINT fk_np FOREIGN KEY (Nurse_id) REFERENCES Nures(Nurse_id);

able altered.

QL> ALTER TABLE patients ADD CONSTRAINT fk_ro FOREIGN KEY (room_no) REFERENCES room(room_no);

able altered.

QL> Insert into patients values (9777,'room',1,'famel','riyadh',051111189, 'm88@outlook.sa',1245, 2209,911);

row created.

QL> Insert into patients values (9119,'hasen',19,'male','makah',08765489, 'ccx8@gmail.sa',1576, 2243,965);

row created.

QL> Insert into patients values (1139,'nada',10,'famel','abha',05611129, 'aas88@outlook.sa',1880, 2876, 950);

row created.

QL> Insert into patients values (9009,'kalid',90,'male','Najran',05836889, 'a388@gmail.sa',1080, 2004,977);

row created.

QL> Insert into patients values (9111,'njla',40,'famel','yunba',01137333, 'a2188@outlook.sa',1139, 2010,0968);

row created.

QL> Insert into patients values (1129,'sultan',28,'male','hail',0561239, '09m88@outlook.sa',1742, 2023,0912);

row created.

QL> Insert into patients values (1219,'nada',10,'famel','abha',05611129, 'aas88@outlook.sa',1623, 2809,0980);

row created.

QL> Insert into patients values (2129,'njla',40,'famel','yunba',01137333, 'a2188@outlook.sa',1980, 2016,0901);

row created.

```

row created.

```
SQL> select * from patients;
```

PATIENT_ID	NAME	AGE	GENDER	ADDRESS	PHONE_NO	EMAIL	DOCTOR_ID	NURSE_ID	ROOM_NO
9777	reem	1	female	riyadh	51111189	m88@outlook.sa	1245	2209	911
9119	hasen	19	male	makah	8765489	ccx8@gmail.sa	1576	2243	965
1139	nada	10	female	abha	5611129	aas88@outlook.sa	1880	2876	950
9009	kalid	90	male	Najran	5836889	a388@gmail.sa	1080	2004	977
9111	njla	40	female	yunba	1137333	a2188@outlook.sa	1139	2010	968
1129	sultan	28	male	hail	561239	09m88@outlook.sa	1742	2023	912
1219	nada	10	female	abha	5611129	aas88@outlook.sa	1623	2809	980
2129	njla	40	female	yunba	1137333	a2188@outlook.sa	1980	2016	901
8976	sultan	28	male	hail	561239	09m88@outlook.sa	1876	2094	951

9 rows selected.

```
SQL> _
```

CREATE AND INSERT RECEPTION

```
create table respcaon ( Reservation_no number (4 ) primary key, Date_entry char (20), Date_exit char (20), Complaints varchar(70), room_no number (4), Patient_id number(4) , Dept_id number(4));
```

```
ALTER TABLE respcaon ADD CONSTRAINT fk_l FOREIGN KEY (room_no) REFERENCES room(room_no);
```

```
ALTER TABLE respcaon ADD CONSTRAINT fk_o FOREIGN KEY (Patient_id) REFERENCES patients (Patient_id);
```

```
ALTER TABLE respcaon ADD CONSTRAINT fk_n FOREIGN KEY (Dept_id ) REFERENCES department (Dept_id );
```

```
insert into respcaon values(6098,'12january','12january','Why is my appointment cancelled',987,2129 ,6266);
```

```
insert into respcaon values(6156,'1february' ,'1february','I want to change the date of the appointment',911,9777,1166);
```

```
insert into respcaon values(6043, '15july','20july','Book an urgent appointment due to complications after the operationd',965,9119,6886);
```

```
insert into respcaon values(6910,'2june','29june','Complaint about the high invoice price',950,1139,4666);
```

```
insert into respcaon values(6977,'8april','8april','Why is my appointment late',977,9009,6066);
```

```
insert into respcaon values(6981,'16august','22august','I could not find the required medicine in the pharmacy',0968,9111,3466);
```

```
insert into respcaon values(6953,'2march','2march','I want to cancel the appointment due to business',0912,1129,3226);
```

```
insert into respcaon values(6970,'17may','1may','I want to make an appointment',0980,2129,3556);
```

```

2
SQL> create table respcaon ( Reservation_no number (4 ) primary key, Date_entry char (20), Date_exit char (20), Complaints
2 varchar(70), room_no number (4), Patient_id number(4) , Dept_id number(4));

Table created.

SQL> ALTER TABLE respcaon ADD CONSTRAINT fk_l FOREIGN KEY (room_no) REFERENCES room(room_no);

Table altered.

SQL> ALTER TABLE respcaon ADD CONSTRAINT fk_o FOREIGN KEY (Patient_id) REFERENCES patients (Patient_id);

Table altered.

SQL> ALTER TABLE respcaon ADD CONSTRAINT fk_n FOREIGN KEY (Dept_id ) REFERENCES department (Dept_id );

Table altered.

SQL> insert into respcaon values(6098,'12january','12january','Why is my appointment cancelled',987,2129 ,6266);

1 row created.

SQL> insert into respcaon values(6156,'1february' , '1february','I want to change the date of the appointment',911,9777,1166);

1 row created.

SQL> insert into respcaon values(6043, '15july','20july','Book an urgent appointment due to complications after the operationd',965,9119,6886);

1 row created.

SQL> insert into respcaon values(6910,'2june','29june','Complaint about the high invoice price',950,1139,4666);

1 row created.

SQL> insert into respcaon values(6977,'8april','8april','Why is my appointment late',977,9009,6066);

1 row created.

SQL> insert into respcaon values(6981,'16august','22august','I could not find the required medicine in the pharmacy',0968,9111,3466);

1 row created.

SQL> insert into respcaon values(6953,'2march','2march','I want to cancel the appointment due to business',0912,1129,3226);

1 row created.

SQL> insert into respcaon values(6970, '17may','1may','I want to make an appointment',0980,2129,3556);

1 row created.

```



```
SQL> select * from respcan;
```

RESERVATION_NO	DATE_ENTRY	DATE_EXIT	COMPLAINTS	ROOM_NO	PATIENT_ID	DEPT_ID
6098	12january	12january	Why is my appointment cancelled	987	2129	6266
6156	1february	1february	I want to change the date of the appointment	911	9777	1166
6043	15july	20july	Book an urgent appointment due to complications after the operationd	965	9119	6886
6910	2june	29june	Complaint about the high invoice price	950	1139	4666
6977	8april	8april	Why is my appointment late	977	9009	6066
6981	16august	22august	I could not find the required medicine in the pharmacy	968	9111	1066
6953	2march	2march	I want to cancel the appointment due to business	912	1129	3226
6970	17may	1may	I want to make an appointment	980	2129	3556

```
8 rows selected.
```

```
SQL>
```

CREATE AND INSERT DEPARTMENT

```
SQL> create table department ( Dept_id number (4 ) primary key, Dept_name varchar (30 ), room_no number ( 4), Patient_id number (4 ), doctor_id number (4 ));
```

```
SQL> ALTER TABLE department ADD CONSTRAINT fk_jk FOREIGN KEY (Patient_id) REFERENCES patients(Patient_id);
```

```
SQL> ALTER TABLE department ADD CONSTRAINT fk_lk FOREIGN KEY (doctor_id ) REFERENCES doctor (doctor_id );
```

```
SQL> ALTER TABLE department ADD CONSTRAINT fk_gh FOREIGN KEY (room_no) REFERENCES room(room_no);
```

```
SQL> Insert into department values (6266, 'dermatophytes', 911, 9777, 1245);
```

```
. SQL> Insert into department values (1166, 'bones', 965, 9119, 1576);
```

```
SQL> Insert into department values (6886,'teeth', 950,1139,1880);
```

```
SQL> Insert into department values (4666,'esotericism', 977,9009,1080);
```

```
SQL> Insert into department values (6066,'suegery',0968,9111,1139);
```

```
SQL> Insert into department values (3466, 'analysis',0912,1129, 1742);
```

```
SQL> Insert into department values (3226, 'heart',0967,1219, 1552);
```

```
SQL> Insert into department values (3556, 'obstetric and gynecology',0901,2129, 1980);
```

```
SQL> Insert into department values (3656, 'eyes',0951,8976, 1876);
```

```

SQL> create table department ( Dept_id number (4 ) primary key, Dept_name varchar (30 ), room_no
2      number ( 4), Patient_id number (4 ), doctor_id number (4 ));
Table created.
SQL> ALTER TABLE department ADD CONSTRAINT fk_jk FOREIGN KEY (Patient_id) REFERENCES patients(Patient_id);
Table altered.
SQL> ALTER TABLE department ADD CONSTRAINT fk_1k FOREIGN KEY (doctor_id ) REFERENCES doctor (doctor_id );
Table altered.
SQL> ALTER TABLE department ADD CONSTRAINT fk_ph FOREIGN KEY (room_no) REFERENCES room(room_no);
Table altered.
SQL> Insert into department values (6266, 'dermatophytes', 911,9777, 1245);
1 row created.
SQL> Insert into department values (1166, 'bones', 965,9119,1576);
1 row created.
SQL> Insert into department values (6886,'teeth', 950,1139,1880);
1 row created.
SQL> Insert into department values (4666,'esotericism', 977,9009,1080);
1 row created.
SQL> Insert into department values (6066,'surgery',0968,9111,1139);
1 row created.
SQL> Insert into department values (3466, 'analysis',0912,1129, 1742);
1 row created.
SQL> Insert into department values (3226, 'heart',0967,1219, 1552);
1 row created.
SQL> Insert into department values (3556, 'obstetric and gynecology',0901,2129, 1980);
1 row created.

```

1 row created.

```
SQL> select * from department;
```

DEPT_ID	DEPT_NAME	ROOM_NO	PATIENT_ID	DOCTOR_ID
6266	dermatophytes	911	9777	1245
1166	bones	965	9119	1576
6886	teeth	950	1139	1880
4666	esotericism	977	9009	1080
6066	suegery	968	9111	1139
3466	analysis	912	1129	1742
3226	heart	967	1219	1552
3556	obstetric and gynecology	901	2129	1980
3656	eyes	951	8976	1876

9 rows selected.

```
SQL> _
```

CREATE AND INSERT DOCTOR

```
create table doctor ( doctor_id number (4 ) primary key , Name varchar (30 ), Age number ( 2), Gender varchar (10 ) , Address varchar ( 30), Phone_no number (10 ) not null unique, Salary number (6 ) not null, Section varchar ( 30 ), Work_hour number (2 ), Email varchar (30 ), Nationality varchar ( 30), Experience number (2 ), Status varchar (10 ), Graduation_uni varchar ( 50) );
```

```
SQL> insert into doctor values(1970,'Faisal',48,'male','Abha',0596882510,68000, 'cardiologist',7,'faisal5@gmail.com','Saudi',16,'married','Stanford University');
```

```
SQL> insert into doctor values(1245,'Jon',32,'male','Riyadh', 0540263482,30000,'dentist',6, 'jon@gmail.com','Italian',6, 'married', 'single University of Oxford');
```

```
SQL> insert into doctor values(1576,'Hind',39,'female','Al- Baha', 0548934511,25000,'ophthalmologist',4, 'hind@gmail.com', 'Italian',10, 'married', 'Univ ersity of King Abdulaziz');
```

```
SQL> insert into doctor values(1880,'Hannen',35,'female','Najran',  
0507148103,40000,'pendiatrician',8,'hanne7nyaser@gmail.com','Jordanian',5,'divorced','University of Cambridge');
```

```
SQL> insert into doctor values(1080,'Khalid',45,'male','Hafr Al Batin', 0502349903,67000,'Psychiatrist',9,  
'kkdl@gmail.com','Saudi',17,'married','University of Hafr Al Batin');
```

```
SQL> insert into doctor values(1139,'Kumaar',50,'male','Tabouk'  
,0590074475,78000,'generalsurgory',9,'kumaarkapur@gmail.com','Indian',21,'married','Harvard University');
```

```
SQL> insert into doctor values(1742,'Sara',38,'female','Qassim', 0532669290,55000,'obsterics gynecology',9  
, 'sara3ahmed@gmail.com','Sudanese',9,'single','Univer sity of King Abdulaziz');
```

```
SQL> insert into doctor  
values(1623,'Maria',37,'female','AlJouf',0554321186,45000,'internist',8,'itisanne@gmail.com','Malaysian',4,'married',  
'National University Bangladesh');
```

```
SQL> insert into doctor values(1552,'Kazuya',44,'male','Al-  
Baha',0548935721,60000,'orthopendic  
secialty',7,'kazuya33@gmail.com','Malaysian',13,'married','University College  
London');
```

```
SQL> insert into doctor values(1980,'Rhaf',31,'female','Al-  
Medina',0507144303,46000,'Anesthesia  
specialist',6,'rrahaf@gmail.com','Saudi',3,'single','Prince Sultan University');
```

```
SQL> insert into doctor values  
(1876,'Ahmed',37,'male','Abha',0522935721,58000,'Dietitian',5,'ahmed@g  
mail.com', 'Saudi',7,'single','King Saud bin Abdulaziz University');
```

```
SQL> insert into doctor values(1456,'Yara',26,'female','Riyadh  
,0505544303,49000,'otolaryngology',5,'yara67@gmail.com','Saudi',2,'married','Pr  
incess Nourah University');
```



```

QL> create table doctor ( doctor_id number (4 ) primary key , Name varchar (30 ) , Age number ( 2), Gender varchar (10 ) , Address varchar ( 30),
  2 Phone_no number (10 ) not null unique, Salary number (6 ) not null, Section varchar ( 30 ) , Work_hour number (2 ) , Email varchar (30 ) ,
  3 Nationality varchar ( 30), Experience number (2 ) , Status varchar (10 ) , Graduation_uni varchar ( 50) );

Table created.

QL> insert into doctor values(1970,'Faisal',48,'male','Abha',0596882510,68000, 'cardiologist',7,'faisal5@gmail.com','Saudi',16,'married','Stanford University');

1 row created.

QL> insert into doctor values(1245,'Jon',32,'male','Riyadh', 0540263482,30000,'dentist',6, 'jon@gmail.com','Italian',6, 'married', 'single University of Oxford');

1 row created.

QL> insert into doctor values(1576,'Hind',39,'female','Al- Baha', 0548934511,25000,'ophthalmologist',4, 'hind@gmail.com', 'Italian',10, 'married', 'Univ ersity of King Abdulaziz');

1 row created.

QL> insert into doctor values(1880,'Hannen',35,'female','Najran', 0507148103,40000,'pendiatrician',8,'hanne7nyaser@gmail.com','Jordanian',5,'divorced','University of Cambridge');

1 row created.

QL> insert into doctor values(1080,'Khalid',45,'male','Hafr Al Batin', 0502349903,67000,'Psychiatrist',9, 'kkdl@gmail.com','Saudi',17,'married','University of Hafr Al Batin');

1 row created.

QL> insert into doctor values(1139,'Kumaar',50,'male','Tabouk ',0590074475,78000,'generalsurgory',9,'kumaarkapur@gmail.com','Indian',21,'married','Harvard University');

1 row created.

QL> insert into doctor values(1742,'Sara',38,'female','Qassim', 0532669290,55000,'obsterics gynecology',9 , 'sara3ahmed@gmail.com','Sudanese',9,'single','Univer sity of King Abdulaziz');

1 row created.

QL> insert into doctor values(1623,'Maria',37,'female','Al- Jouf',0554321186,45000,'internist',8,'itisanne@gmail.com','Malaysian',4,'married', 'National University Bangladesh');

1 row created.

QL> insert into doctor values(1552,'Kazuya',44,'male','Al-Baha',0548935721,60000,'orthpendic secialty',7,'kazuya33@gmail.com','Malaysian',13,'married','University College London');

1 row created.

QL> insert into doctor values(1980,'Rhaf',31,'female','Al-Medina',0507144303,46000,'Anesthesia specialist',6,'rrahaf@gmail.com','Saudi',3,'single','Prince Sultan University');

1 row created.

QL> insert into doctor values(1876,'Ahmed',37,'male','Abha',0522935721,58000,'Dietitian',5,'ahmed@g mail.com', 'Saudi',7,'single','King Saud bin Abdulaziz University');

1 row created.

```

```
SQL> select * from doctor;
```

DOCTOR_ID	NAME	EXPERIENCE	STATUS	GRADUATION_UNI	AGE	GENDER	ADDRESS	PHONE_NO	SALARY	SECTION	WORK_HOUR	EMAIL	NATIONALITY
1970	Faisal	16	married	Stanford University	48	male	Abha	596882510	68000	cardiologist	7	faisal5@gmail.com	Saudi
1245	Jon	6	married	single University of Oxford	32	male	Riyadh	540263482	30000	dentist	6	jon@gmail.com	Italian
1576	Hind	10	married	Univ ersity of King Abdulaziz	39	female	Al- Baha	548934511	25000	ophthalmologist	4	hind@gmail.com	Italian
1880	Hannen	5	divorced	University of Cambridge	35	female	Najran	507148103	40000	pendiatrician	8	hanne7nyaser@gmail.com	Jordanian
1080	Khalid	17	married	University of Hafr Al Batin	45	male	Hafr Al Batin	502349903	67000	Psychiatrist	9	kkdl@gmail.com	Saudi
1139	Kumaar	21	married	Harvard University	50	male	Tabouk	590074475	78000	generalsurgory	9	kumaarkapur@gmail.com	Indian
1742	Sara	9	single	Univer sity of King Abdulaziz	38	female	Qassim	532669290	55000	obsterics gynecology	9	sara3ahmed@gmail.com	Sudanese
1623	Maria	4	married	National University Bangladesh	37	female	Al- Jouf	554321186	45000	internist	8	itisanne@gmail.com	Malaysian
1552	Kazuya	13	married	University College London	44	male	Al-Baha	548935721	60000	orthpendic secialty	7	kazuya33@gmail.com	Malaysian
1980	Rhaf	3	single	Prince Sultan University	31	female	Al-Medina	507144303	46000	Anesthesia specialist	6	rrahaf@gmail.com	Saudi
1876	Ahmed	7	single	King Saud bin Abdulaziz University	37	male	Abha	522935721	58000	Dietitian	5	ahmed@e mail.com	Saudi
1456	Yara	2	married	Pr incess Mourah University	26	female	Riyadh	505544303	49000	otolaryngology	5	yara67@gmail.com	Saudi

12 rows selected.

CREATE AND INSERT NURSE

```
SQL> create table Nures ( Nurse_id number (4 ) primary key, Name varchar (30 ), Age number (2 ), Gender varchar (10 ), Address varchar ( 30), Phone_no number (10 ) not null, Salary number ( 6) not null, Work_hour number (2 ), Email varchar (30 ) , Nationality varchar ( 30), Experience number ( 2), Status varchar (10 ), Graduation_uni varchar ( 30) );
```

```
SQL> insert into Nures  
values(2234,'Abrar',35,'female','ABha',0567782998,35000,7,'abrar66@gmail.com','Saudi',7,'married','Princess Nourah University');
```

```
SQL> insert into Nures  
values(2209,'Sumaya',31,'female','Riyadh',0569002998,20000,5,'suma@gmail.com','Saudi',5,'married','King Khalid University');
```

```
SQL> insert into Nures  
values(2243,'zahra',25,'female','Najran',0512302998,26000,8,'zaha9@gmail.com','Saudi',3,'single','University of King Abdulaziz');
```

```
SQL> insert into Nures  
values(2876,'Afnan',34,'female','Qassim',0512365431,23000,9,'afnoon@gmail.com','Saudi',8,'married','University of Hafr  
Al Batin');
```

```
SQL> insert into Nures  
values(2004,'Amirah',29,'female','Tabouk',0560701298,48000,6,'amirah@gmail.com','Saudi',5,'married','University College  
London');
```

```
SQL> insert into Nures  
values(2010,'Wejdan',42,'female','Bisha',0588431291,59000,8,'wajod@gmail.com','Saudi',18,'married','Jouf University');
```

```
SQL> insert into Nures  
values(2023,'Renad',23,'female','Dammam',0582431861,20900,7,'ra09@gmail.com','Saudi',1,'single','Prince Sultan  
University');
```

```
SQL> insert into Nures  
values(2809,'Maram',37,'female','AlHareeq',058288643,28000,6,'marr8@gmail.com','Saudi',8  
, 'married', 'Princess Nourah University');
```

```
SQL> insert into Nures values(2870,'Zinab',26,'female','Khamis  
Mushayt',058288876,37000,7,'znabo@gmail.com','Saudi',4,'single','Al-Yamamah University');
```

```
SQL> insert into Nures  
values(2016,'Jumanah',30,'female','Qatif',058205476,44000,9,'juj09@gmail.com','Saudi',6,'single',  
'University of Prince Mugrin');
```

```
SQL> insert into Nures  
values(2094,'Nadaas',38,'female','Yanbu',0587665431,18000,9,'anado8@gmail.com','Saudi',4,'  
married','King Saud University');
```

```
SQL> insert into Nures  
values(2074,'Rawan',38,'female','Tabouk',0560700198,37000,6,'rawanmoha@gmail.com','Saudi',  
'12','married','National Medical Studies');
```

```

SQL> create table Nures ( Nurse_id number (4 ) primary key, Name varchar (30 ), Age number (2 ), Gender varchar (10 ),
2 Address varchar (30), Phone_no number (10 ) not null, Salary number (6) not null, Work_hour number (2 ), Email varchar (30 ), Nationality varchar (30), Experience number (2), Status varchar (10)
Graduation_uni varchar (30) );

Table created.

SQL> insert into Nures values(2234,'Abrar',35,'female','Abha',0567782998,35000,7,'abrar66@gmail.com','Saudi',7,'married','Princess Nourah University');

1 row created.

SQL> insert into Nures values(2209,'Sumaya',31,'female','Riyadh',0569082998,20000,5,'suma@gmail.com','Saudi',5,'married','King Khalid University');

1 row created.

SQL> insert into Nures values(2243,'zahra',25,'female','Najran',0512302998,26000,8,'zaha9@gmail.com','Saudi',3,'single','University of King Abdulaziz');

1 row created.

SQL> insert into Nures values(2876,'Afnan',34,'female','Qassim',0512365431,23000,9,'afnoon@gmail.com','Saudi',8,'married','University of Hafr Al Batin');

1 row created.

SQL> insert into Nures values(2004,'Amirah',29,'female','Tabouk',0560701298,48000,6,'amirah@gmail.com','Saudi',5,'married','University College London');

1 row created.

SQL> insert into Nures values(2030,'Wejdan',42,'female','Bisha',0588431291,59000,8,'wajod@gmail.com','Saudi',18,'married','Jouf University');

1 row created.

SQL> insert into Nures values(2023,'Renad',23,'female','Dammam',0582431861,20900,7,'ra09@gmail.com','Saudi',1,'single','Prince Sultan University');

1 row created.

SQL> insert into Nures values(2809,'Maram',37,'female','Al-Harreeq',058288643,28000,6,'marr8@gmail.com','Saudi',8,'married','Princess Nourah University');

1 row created.

SQL> insert into Nures values(2870,'Zinah',26,'female','Khanis Mushayt',058288876,37000,7,'znabo@gmail.com','Saudi',4,'single','Al-Yamamah University');

1 row created.

SQL> insert into Nures values(2016,'Jumanah',30,'female','Qatif',058205476,44000,9,'juj99@gmail.com','Saudi',6,'single','University of Prince Muqrin');

1 row created.

SQL> insert into Nures values(2094,'Nadaas',38,'female','Yanbu',0587665431,18000,9,'anado8@gmail.com','Saudi',4,'married','King Saud University');

1 row created.

```


NURSE_ID	NAME	AGE	GENDER	ADDRESS	PHONE_NO	SALARY	WORK_HOUR	EMAIL	NATIONALITY	EXPERIENCE	STATUS
GRADUATION_UNI											
2234	Abrar	35	female	ABha	567782998	35000	7	abrar66@gmail.com	Saudi	7	married
Princess Nourah University											
2209	Sumaya	31	female	Riyadh	569002998	20000	5	sumo@gmail.com	Saudi	5	married
King Khalid University											
2243	zahra	25	female	Najran	512302998	26000	8	zaha9@gmail.com	Saudi	3	single
University of King Abdulaziz											
2876	Afnan	34	female	Qassim	512365431	23000	9	afnoon@gmail.com	Saudi	8	married
University of Hafr Al Batin											
2004	Amirah	29	female	Tabouk	560701298	48000	6	amirah@gmail.com	Saudi	5	married
University College London											
2010	Wejdan	42	female	Bisha	588431291	59000	8	wejod@gmail.com	Saudi	18	married
Jouf University											
2023	Renad	23	female	Dammam	582431861	20900	7	re09@gmail.com	Saudi	1	single
Prince Sultan University											
2809	Maram	37	female	Al-Hareeq	58288643	28000	6	marr8@gmail.com	Saudi	8	married
Princess Nourah University											
2870	Zinab	26	female	Khamis Mushayt	58288876	37000	7	znabo@gmail.com	Saudi	4	single
Al-Yamamah University											
2016	Jumanah	30	female	Qatif	58205476	44000	9	juj09@gmail.com	Saudi	6	single
University of Prince Mugrin											
2094	Nadaas	38	female	Yanbu	587665431	18000	9	anado8@gmail.com	Saudi	4	married
King Saud University											
NURSE_ID	NAME	AGE	GENDER	ADDRESS	PHONE_NO	SALARY	WORK_HOUR	EMAIL	NATIONALITY	EXPERIENCE	STATUS
GRADUATION_UNI											
2074	Rawan	38	female	Tabouk	560700198	37000	6	rawannoha@gmail.com	Saudi	12	married
National Medical Studies											

CREATE AND INSERT ROOM

```
SQL> create table room ( room_no number ( 4) primary key, room_type varchar ( 30), situation varchar (30 ), facilities varchar ( 30));
```

```
SQL> insert into Room values(987,'wing degree','reserved','Nada');
```

```
SQL> insert into Room values(911,'Beds emergency service','empty','Saleh');
```

```
SQL> insert into Room values(965,'Excellent first class','Full of','Ali');
```

```
SQL> insert into Room values(950,'Beds emergency service','reserved','Abrar');
```

```
SQL> insert into Room values(977,'Excellent first class','service','Fatimah');
```



```
SQL> insert into Room values(0968,'wing degree','Full of','Fatimah');
```

```
SQL> insert into Room values(0912,'Deluxe First Class','reserved','Jumanah');
```

```
SQL> insert into Room values(0980,'Beds emergency service','empty','Gadah');
```

```
SQL> insert into Room values(0967,'Excellent first class ','reserved','Hannen');
```

```
SQL> insert into Room values(0901,'Beds emergency service','Full of','Noha');
```

```
SQL> insert into Room values(0951,'Deluxe First Class','service','Yara');
```

```
SQL> insert into Room values(0970,'wing degree','reserved','Fzilah');
```

```
SQL> create table room ( room_no number ( 4) primary key, room_type varchar ( 30), situation varchar (30 ),  
2 facilities varchar ( 30));
```

Table created.

```
SQL> insert into Room values(987,'wing degree','reserved','Nada');
```

1 row created.

```
SQL> insert into Room values(911,'Beds emergency service','empty','Saleh');
```

1 row created.

```
SQL> insert into Room values(965,'Excellent first class','Full of','Ali');
```

1 row created.

```
SQL> insert into Room values(950,'Beds emergency service','reserved','Abrar');
```

1 row created.

```
SQL> insert into Room values(977,'Excellent first class','service','Fatimah');
```

1 row created.

```
SQL> insert into Room values(0968,'wing degree','Full of','Fatimah');
```

1 row created.

```
SQL> insert into Room values(0912,'Deluxe First Class','reserved','Jumanah');
```

1 row created.

```
SQL> insert into Room values(0980,'Beds emergency service','empty','Gadah');
```

1 row created.

```
SQL> insert into Room values(0967,'Excellent first class ','reserved','Hannen');
```

1 row created.

```
SQL> insert into Room values(0901,'Beds emergency service','Full of','Noha');
```

1 row created.

```
SQL> insert into Room values(0951,'Deluxe First Class','service','Yara');
```

1 row created.

1 row created.

```
SQL> select * from room;
```

ROOM_NO	ROOM_TYPE	SITUATION	FACILITIES
987	wing degree	reserved	Nada
911	Beds emergency service	empty	Saleh
965	Excellent first class	Full of	Ali
950	Beds emergency service	reserved	Abrar
977	Excellent first class	service	Fatimah
968	wing degree	Full of	Fatimah
912	Deluxe First Class	reserved	Jumanah
980	Beds emergency service	empty	Gadah
967	Excellent first class	reserved	Hannen
901	Beds emergency service	Full of	Noha
951	Deluxe First Class	service	Yara
970	wing degree	reserved	Fzilah

12 rows selected.

```
SQL>
```

CREATE AND INSERT BILL

```
create table bill (bill_no number (10 ) primary key, Patient_id number (4 ) , doctor_id number (4), room_no number ( 4 ),days_no number (10), Dept_id number (4) , bill number ( 10));
```

```
SQL> ALTER TABLE Bill ADD CONSTRAINT fk_q FOREIGN KEY (doctor_id) REFERENCES doctor (doctor_id);
```

```
SQL> ALTER TABLE Bill ADD CONSTRAINT fk_g FOREIGN KEY (room_no) REFERENCES room(room_no) 2 ;
```

```
SQL> ALTER TABLE Bill ADD CONSTRAINT fk_j FOREIGN KEY (Patient_id) REFERENCES patients(Patient_id) ;
```

```
SQL> Insert into Bill values (2787, 8976, 1970 , 987 , 2 , 3226 , 150);
```

```
SQL> Insert into Bill values (7387, 9777 , 1245 , 911 , 1, 6266 , 1242);
```

SQL> Insert into Bill values (5787, 9119, 1576 , 965 , 6 , 1166 , 350);

SQL> Insert into Bill values (8787, 1139, 1880 , 950 , 30 , 6886, 250);

SQL> Insert into Bill values (7087, 9009, 1080 , 977 , 2 , 4666 , 350);

SQL> Insert into Bill values (7287, 9111, 1139 , 0968 , 7, 6066 , 850);

SQL> Insert into Bill values (7727, 1129, 1742 , 0912 , 15 , 3466 , 1520);

SQL> Insert into Bill values (7487, 1129, 1552 , 0967 , 9, 3556 , 1350);

SQL> Insert into Bill values (7587, 2129, 1980 , 0901 , null , 3656, 167);

Table altered.

```
SQL> ALTER TABLE Bill ADD CONSTRAINT fk_g FOREIGN KEY (room_no) REFERENCES room(room_no)
      2 2 ;
      2
      *
```

ERROR at line 2:

ORA-01735: invalid ALTER TABLE option

```
SQL> ALTER TABLE Bill ADD CONSTRAINT fk_g FOREIGN KEY (room_no) REFERENCES room(room_no) 2 ;
      ALTER TABLE Bill ADD CONSTRAINT fk_g FOREIGN KEY (room_no) REFERENCES room(room_no) 2
      *
```

ERROR at line 1:

ORA-01735: invalid ALTER TABLE option

```
SQL> ALTER TABLE Bill ADD CONSTRAINT fk_g FOREIGN KEY (room_no) REFERENCES room(room_no);
```

Table altered.

```
SQL> ALTER TABLE Bill ADD CONSTRAINT fk_j FOREIGN KEY (Patient_id) REFERENCES patients(Patient_id) ;
```

Table altered.

```
SQL> Insert into Bill values (2787, 8976,1970 , 987 , 2 , 3226 ,150);
```

1 row created.

```
SQL> Insert into Bill values (7387,9777 ,1245 , 911 , 1, 6266 ,1242);
```

1 row created.

```
SQL> Insert into Bill values (5787, 9119,1576 , 965 , 6 , 1166 ,350);
```

1 row created.

```
SQL> Insert into Bill values (8787, 1139,1880 ,950 , 30 , 6886,250);
```

1 row created.

1 row created.

```
SQL> Insert into Bill values (8787, 1139,1880 ,950 , 30 , 6886,250);
```

1 row created.

```
SQL> Insert into Bill values (7087, 9009,1080 , 977 , 2 , 4666 ,350);
```

1 row created.

```
SQL> Insert into Bill values (7287, 9111,1139 ,0968 , 7, 6066 ,850);
```

1 row created.

```
SQL> Insert into Bill values (7727, 1129,1742 ,0912 , 15 , 3466 ,1520);
```

1 row created.

```
SQL> Insert into Bill values (7487, 1129,1552 ,0967 , 9, 3556 ,1350);
```

1 row created.

```
SQL> Insert into Bill values (7587, 2129,1980 ,0901 , null , 3656,167);
```

1 row created.

```
SQL> select * from bill;
```

BILL_NO	PATIENT_ID	DOCTOR_ID	ROOM_NO	DAYS_NO	DEPT_ID	BILL
2787	8976	1970	987	2	3226	150
7387	9777	1245	911	1	6266	1242
5787	9119	1576	965	6	1166	350
8787	1139	1880	950	30	6886	250
7087	9009	1080	977	2	4666	350
7287	9111	1139	968	7	6066	850
7727	1129	1742	912	15	3466	1520
7487	1129	1552	967	9	3556	1350
7587	2129	1980	901		3656	167

9 rows selected.

```
SQL> _
```

THE SQL QUERIES

- `select n.name , n.age ,Administrat_id, a.gender from nures n full outer join Administration a on n.nurse_id = a.Administrat_id;`
- `select n.name , n.age ,Administrat_id, a.gender from nures n left outer join Administration a on n.nurse_id = a.Administrat_id;`
- `select S.jop_title , S.name , S.age ,Administrat_id, a.gender from stuff s right outer join Administration a on s.Stuff_id = a.Administrat_id;`
- `select n.name , n.age ,Administrat_id, a.gender , a.salary from nures n inner join Administration a on a.salary>1500;`


```
SQL> select  n.name , n.age , Administrat_id, a.gender from nures n full outer join Administration a on n.nurse_id = a.Administrat_id;
```

NAME	AGE	ADMINISTRAT_ID	GENDER
Abrar	35		
Sumaya	31		
zohra	25		
Afnan	34		
Amirah	29		
Wejdan	42		
Renad	23		
Maram	37		
Zinab	26		
Jumanah	30		
Nadaas	38		

NAME	AGE	ADMINISTRAT_ID	GENDER
Rawan	38		
		5123	male
		5419	female
		5600	female
		5098	female
		5437	female
		5567	male
		5020	male
		5451	female
		5117	male
		5166	male

22 rows selected.

```
SQL> █
```

NAME	AGE	ADMINISTRAT_ID	GENDER
Abrar	35		
Dumanah	30		
Rowan	38		
Renad	23		
Nadaas	38		
Maran	37		
Afnan	34		
Mejdan	42		
Sumaya	31		
Zinab	26		
Anirah	29		

NAME	AGE	ADMINISTRAT_ID	GENDER
zahra	25		

12 rows selected.

SQL>

```
SQL> select s.job_title , s.name , s.age , Administrat_id, a.gender from stuff s right outer join Administration a on s.Stuff_id = a.Administrat_id;
```

JOB_TITLE	NAME	AGE	ADMINISTRAT_ID	GENDER
			5123	male
			5419	female
			5600	female
			5098	female
			5437	female
			5567	male
			5020	male
			5451	female
			5117	male
			5166	male

10 rows selected.

```
SQL>
```

```
SQL> select n.name , n.age , Administrat_id, a.gender , a.salary from nures n inner join Administration a on a.salary>1500;
```

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Abnar	35	5098	female	12000
Sumaya	31	5098	female	12000
zahra	25	5098	female	12000
Afnan	34	5098	female	12000
Amirah	29	5098	female	12000
wejdani	42	5098	female	12000
Renad	23	5098	female	12000
Maran	37	5098	female	12000
Zinab	26	5098	female	12000
Jumanah	30	5098	female	12000
Nadaas	38	5098	female	12000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Rawan	38	5098	female	12000
Abnar	35	5123	male	10000
Sumaya	31	5123	male	10000
zahra	25	5123	male	10000
Afnan	34	5123	male	10000
Amirah	29	5123	male	10000
wejdani	42	5123	male	10000
Renad	23	5123	male	10000
Maran	37	5123	male	10000
Zinab	26	5123	male	10000
Jumanah	30	5123	male	10000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Nadaas	38	5123	male	10000
Rawan	38	5123	male	10000
Abnar	35	5166	male	20000
Sumaya	31	5166	male	20000
zahra	25	5166	male	20000
Afnan	34	5166	male	20000
Amirah	29	5166	male	20000
wejdani	42	5166	male	20000
Renad	23	5166	male	20000
Maran	37	5166	male	20000
Zinab	26	5166	male	20000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Jumanah	30	5166	male	20000
Nadaas	38	5166	male	20000
Rawan	38	5166	male	20000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Jumanah	38	5166	male	20000
Nadaas	38	5166	male	20000
Rawan	38	5166	male	20000
Abrar	35	5567	male	9000
Sumaya	31	5567	male	9000
zahra	25	5567	male	9000
Afnan	34	5567	male	9000
Asirah	29	5567	male	9000
Wejdan	42	5567	male	9000
Renad	23	5567	male	9000
Maran	37	5567	male	9000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Zinab	26	5567	male	9000
Jumanah	38	5567	male	9000
Nadaas	38	5567	male	9000
Rawan	38	5567	male	9000
Abrar	35	5437	female	15000
Sumaya	31	5437	female	15000
zahra	25	5437	female	15000
Afnan	34	5437	female	15000
Asirah	29	5437	female	15000
Wejdan	42	5437	female	15000
Renad	23	5437	female	15000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Maran	37	5437	female	15000
Zinab	26	5437	female	15000
Jumanah	38	5437	female	15000
Nadaas	38	5437	female	15000
Rawan	38	5437	female	15000
Abrar	35	5020	male	25000
Sumaya	31	5020	male	25000
zahra	25	5020	male	25000
Afnan	34	5020	male	25000
Asirah	29	5020	male	25000
Wejdan	42	5020	male	25000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Renad	23	5020	male	25000
Maran	37	5020	male	25000
Zinab	26	5020	male	25000
Jumanah	38	5020	male	25000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Jumanah	30	5166	male	20000
Nadaas	38	5166	male	20000
Rawan	38	5166	male	20000
Abnar	35	5567	male	9000
Sumaya	31	5567	male	9000
zahra	25	5567	male	9000
Afnan	34	5567	male	9000
Asirah	29	5567	male	9000
dejdan	42	5567	male	9000
Renad	23	5567	male	9000
Naran	37	5567	male	9000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Zinab	26	5567	male	9000
Jumanah	30	5567	male	9000
Nadaas	38	5567	male	9000
Rawan	38	5567	male	9000
Abnar	35	5437	female	15000
Sumaya	31	5437	female	15000
zahra	25	5437	female	15000
Afnan	34	5437	female	15000
Asirah	29	5437	female	15000
dejdan	42	5437	female	15000
Renad	23	5437	female	15000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Naran	37	5437	female	15000
Zinab	26	5437	female	15000
Jumanah	30	5437	female	15000
Nadaas	38	5437	female	15000
Rawan	38	5437	female	15000
Abnar	35	5020	male	25000
Sumaya	31	5020	male	25000
zahra	25	5020	male	25000
Afnan	34	5020	male	25000
Asirah	29	5020	male	25000
dejdan	42	5020	male	25000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Renad	23	5020	male	25000
Naran	37	5020	male	25000
Zinab	26	5020	male	25000
Jumanah	30	5020	male	25000
Nadaas	38	5020	male	25000

Renad	23	5020	male	25000
Maran	37	5020	male	25000
Zinab	26	5020	male	25000
Jumanah	30	5020	male	25000
Nadaas	38	5020	male	25000
Rawan	38	5020	male	25000
Abrar	35	5419	female	13000
Sumaya	31	5419	female	13000
zahra	25	5419	female	13000
Afnan	34	5419	female	13000
Amirah	29	5419	female	13000
NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Wejdan	42	5419	female	13000
Renad	23	5419	female	13000
Maran	37	5419	female	13000
Zinab	26	5419	female	13000
Jumanah	30	5419	female	13000
Nadaas	38	5419	female	13000
Rawan	38	5419	female	13000
Abrar	35	5451	female	31000
Sumaya	31	5451	female	31000
zahra	25	5451	female	31000
Afnan	34	5451	female	31000
NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Amirah	29	5451	female	31000
Wejdan	42	5451	female	31000
Renad	23	5451	female	31000
Maran	37	5451	female	31000
Zinab	26	5451	female	31000
Jumanah	30	5451	female	31000
Nadaas	38	5451	female	31000
Rawan	38	5451	female	31000
Abrar	35	5600	female	25000
Sumaya	31	5600	female	25000
zahra	25	5600	female	25000
NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
Afnan	34	5600	female	25000
Amirah	29	5600	female	25000
Wejdan	42	5600	female	25000
Renad	23	5600	female	25000
Maran	37	5600	female	25000
Zinab	26	5600	female	25000
Jumanah	30	5600	female	25000

Renad	23	5451	female	31000
Maran	37	5451	female	31000
Zinab	26	5451	female	31000
Jumanah	30	5451	female	31000
Nadaas	38	5451	female	31000
Rawan	38	5451	female	31000
Abrar	35	5600	female	25000
Sumaya	31	5600	female	25000
zahra	25	5600	female	25000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
------	-----	----------------	--------	--------

Afnan	34	5600	female	25000
Amirah	29	5600	female	25000
Wajdan	42	5600	female	25000
Renad	23	5600	female	25000
Maran	37	5600	female	25000
Zinab	26	5600	female	25000
Jumanah	30	5600	female	25000
Nadaas	38	5600	female	25000
Rawan	38	5600	female	25000
Abrar	35	5117	male	14000
Sumaya	31	5117	male	14000

NAME	AGE	ADMINISTRAT_ID	GENDER	SALARY
------	-----	----------------	--------	--------

zahra	25	5117	male	14000
Afnan	34	5117	male	14000
Amirah	29	5117	male	14000
Wajdan	42	5117	male	14000
Renad	23	5117	male	14000
Maran	37	5117	male	14000
Zinab	26	5117	male	14000
Jumanah	30	5117	male	14000
Nadaas	38	5117	male	14000
Rawan	38	5117	male	14000

120 rows selected.

THE REFERENCES

- link1: <https://images.app.goo.gl/PSEeNFufTgE4GZkU8>
- link2: <https://images.app.goo.gl/tVthFcsZQKXRf4JH7>
- link3: <https://www.w3resource.com/sql-exercises/hospitaldatabase-exercise/index.php>
- link4: <https://itsourcecode.com/free-projects/vb-net/hospitalmanagement-system-database-design/>
- link5: <https://www.db-book.com/db6/slide-dir/>

THE END